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Original Article

A comprehensive evaluation of multicentric reliability of single-subject cortical morphological networks on traveling subjects

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Despite the prevalence of research on single-subject cerebral morphological networks in recent years, whether they can offer a reliable way for multicentric studies remains largely unknown. Using two multicentric datasets of traveling subjects, this work systematically examined the inter-site test-retest (TRT) reliabilities of single-subject cerebral morphological networks, and further evaluated the effects of several key factors. We found that most graph-based network measures exhibited fair to excellent reliabilities regardless of different analytical pipelines. Nevertheless, the reliabilities were affected by choices of morphological index (fractal dimension > sulcal depth > gyrification index > cortical thickness), brain parcellation (high-resolution > low-resolution), thresholding method (proportional > absolute), and network type (binarized > weighted). For the factor of similarity measure, its effects depended on the thresholding method used (absolute: Kullback–Leibler divergence > Jensen–Shannon divergence; proportional: Jensen–Shannon divergence > Kullback–Leibler divergence). Furthermore, longer data acquisition intervals and different scanner software versions significantly reduced the reliabilities. Finally, we showed that inter-site reliabilities were significantly lower than intra-site reliabilities for single-subject cerebral morphological networks. Altogether, our findings propose single-subject cerebral morphological networks as a promising approach for multicentric human connectome studies, and offer recommendations on how to determine analytical pipelines and scanning protocols for obtaining reliable results.

Key words: morphological brain network; structural MRI; multicentric study; test-retest reliability; graph theory.

Introduction

Morphological brain networks encode interregional relations in regional morphology derived from structural magnetic resonance imaging (MRI). Traditionally, morphological brain networks are constructed by calculating correlations of certain morphological features [e.g. gray matter volume, cortical thickness (CT), or surface area] between brain regions across a cohort of participants (He et al. 2007; Bassett et al. 2008; Sanabria-Diaz et al. 2010). The population-based approaches, however, is powerless for individualized clinical diagnosis and assessment due to the neglect of interindividual differences. To respond, several approaches have recently been developed to construct morphological brain networks at the individual level (Tijms et al. 2012; Kong et al. 2015; Wang et al. 2016; Jiang et al. 2017; Li et al. 2017; Yu et al. 2018; Li et al. 2021).Currently, individual-level single-subject morphological brain networks have been becoming an important way to explore human connectome in health and disease.

A distinct advantage of single-subject morphological brain networks is their consistently reported high test-retest (TRT) reliabilities in topologically characterizing organizational principles that govern wiring patterns of the human brain (Tijms et al. 2012; Kong et al. 2015; Wang et al. 2016; Jiang et al. 2017; Li et al. 2017; Yu et al. 2018; Li et al. 2021). This feature makes single-subject morphological brain networks promising in identifying reliable diagnostic and prognostic biomarkers in clinic. For instance, previous studies have shown that single-subject morphological brain networks are able to classify patients with transient ischemic attack from controls (Lv et al. 2021), and predict the progression from mild cognitive impairment to Alzheimer's disease (Zheng et al. 2018). However, it should be noted that current findings of the high TRT reliabilities of single-subject morphological brain networks are all based on scan-rescans collected in a single center or site.

In recent years, multicentric studies have attracted considerable attention due to their advantages of larger sample size, lower data collection costs, increased statistical power, and higher generalizability. A prerequisite for such studies is the reproducibility of data collected in different sites, which, however, is increasingly reported to be affected by geography, models, hardware and software, and other factors (Mirzaalian et al. 2016; Fortin et al. 2017; Tong et al. 2019; Voelker et al. 2021; Hedges et al. 2022). With respect to topological organization of large-scale brain networks, a recent study found that different scanners led to lower TRT reliabilities of graph-based network measures for diffusion

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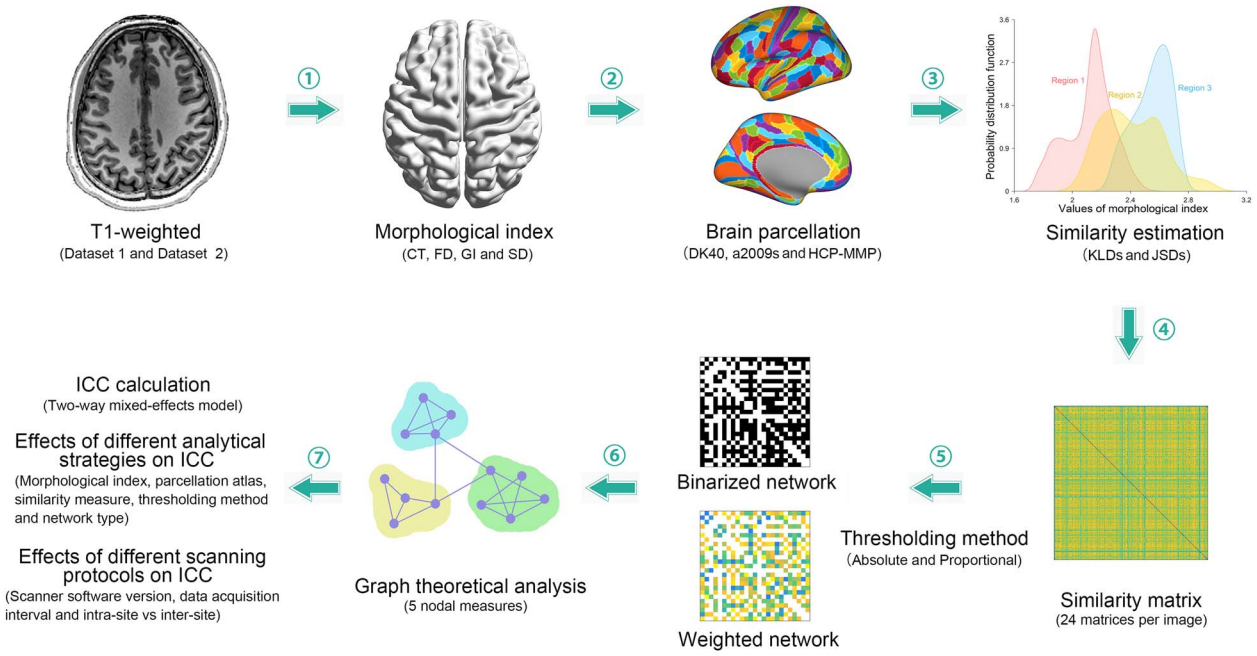


Fig. 1. Flow chart of the entire analytical pipeline. ① For each structural image, four vertexwise morphological maps were first obtained. ② Each morphological map was then separately divided into different regions according to three prior cortical surface atlases. ③ Subsequently, values of each morphological index were extracted from each region to estimate regional probability distribution functions. ④ The probability distribution functions were further used to calculate interregional morphological similarity based on two similarity measures, resulting in a total of 24 morphological similarity matrices for each structural image. ⑤ Afterwards, two thresholding methods were separately used to convert each morphological similarity matrix to a series of binarized and weighted networks. ⑥ To topologically characterize the networks, five nodal graph-based network measures were calculated. ⑦ Finally, the inter-site TRT reliabilities were quantified for each network measure using the intra-class correlation coefficient, and used to examine the effects of different analytical strategies and scanning protocols. CT, cortical thickness; FD, fractal dimension; GI, gyrification index; SD, sulcal depth; HCP-MMP, Human Connectome Project Multi-Modal Parcellation; KLDs, Kullback–Leibler divergence-based similarity; JSDs, Jensen–Shannon divergence-based similarity; ICC, intra-class correlation coefficient.

MRI-based structural brain networks even when matched protocols were used (Kurokawa et al. 2021). To date, there are no studies evaluating the TRT reliabilities of single-subject morphological brain networks across multiple sites.

In this study, we used two multicentric datasets of traveling subjects to systematically investigate the TRT reliabilities of single-subject morphological brain networks. Specifically, we constructed four types of single-subject cerebral morphological brain networks, and evaluated the effects of several key factors on the inter-site TRT reliabilities of the networks, including morphological index, brain parcellation, similarity measure, thresholding method, network type, data acquisition interval, and scanner software version. In addition, the inter-site TRT reliabilities were compared with the reliabilities derived from scan-rescans collected in a single site. We hypothesize that the inter-site TRT reliabilities of single-subject cerebral morphological networks are high but depend on the choices of different analytical strategies and scanning protocols.

Materials and methods

General descriptions of analytical pipeline

In this study, we analyzed T1-weighted structural MRI images from two publicly available datasets: the first dataset (dataset 1) included three healthy participants scanned in ten sites (Tong et al. 2019), and the second one (dataset 2) included nine participants scanned in eleven sites (Tanaka et al. 2021). First, for each structural MRI image in the two datasets, we derived four surface-based, vertexwise morphological brain maps: CT, fractal dimension (FD), gyrification index (GI), and sulcal depth (SD). The four morphological indices were chosen because (i) they are widely used to explore cortical architecture in brain development, aging, and disease; (ii) they offer nonredundant descriptions of cortical characteristics that exhibit differential developmental and aging trajectories and are differently affected in various brain diseases; and (iii) they are all computationally available from the CAT12 toolbox, which offers a volume-based approach for estimating cerebral surface morphology without extensive reconstruction of cortical surface and thus is timesaving. Then, six single-subject morphological similarity matrices were constructed for each morphological brain map using three cortical surface atlases for brain parcellation (68, 148, and 360 regions) and two measures for estimating interregional morphological similarity (Kullback–Leibler divergence-based similarity, KLDs, and Jensen–Shannon divergence-based similarity, JSDs). These morphological similarity matrices were further thresholded using two thresholding procedures (absolute and proportional) to generate a set of both binarized and weighted cerebral morphological networks. Afterwards, several graph-based network measures were calculated for each region to topologically characterize their roles in the cerebral morphological networks. Finally, the intra-class correlation coefficient (ICC) was used to quantify the inter-site TRT reliabilities of each network measure of each region, followed by examinations of the effects of morphological index, brain parcellation, similarity measure, thresholding method, and network type on the reliabilities. In addition, for dataset 1, we further examined the effects of data acquisition interval and scanner software version on the reliabilities, and compared the reliabilities with those derived from intra-site scan-rescans. Fig. 1 illustrates a flowchart of the entire data analytical pipeline in this study.

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Participants and data acquisition

Dataset 1

Three healthy traveling subjects (one male, 23 years old, and two females, 26 and 23 years old, respectively) were included in dataset 1 (Tong et al. 2019), who were scanned once in nine centers (termed as Center 1–Center 9), and three times in the other center (termed as Center 10–1, Center 10–2 and Center 10–3). All scans were performed by the same operator according to the fixed operation process in 3 T MR MAGNETOM Prisma (Siemens, Erlangen, Germany), equipped with max gradient strength of 80 mT/m and slew rate of 200 T/m/s. The scanner software version was Syngo MR VD13D for all scanners except for the scanner in the Center 9 (Syngo MR VE11C). For all scans, the same type of 64-channel head coil and the same acquisition parameters were used. Specifically, all structural MRI images were acquired using a 3D magnetization-prepared two rapid acquisition gradient echo (MP2RAGE) sequence with the following imaging parameters: repetition time/echo time = 5 s/2.9 ms, inversion time = 700, 2,500 ms, field of view = 211 × 256 × 256 mm^3, voxel size = 1.2 × 1 × 1 mm^3, bandwidth = 240 Hz/Px, generalized auto-calibrating partial parallel acquisition factor = 3, and the acquisition time was 8 minutes and 22 seconds.

Dataset 2

Nine healthy traveling subjects (all males; age range 24–32 years old; mean age 27 ± 2.6 years) were included in dataset 2 (Tanaka et al. 2021), who were scanned once in 10 centers, and five cycles (morning, afternoon, the following day, the following week, and the following month) in the other center (termed as ATT). Key imaging parameters used at each center for the T1-weighted structural MRI images are summarized in Table S1.

Processing of structural MRI data

Preprocessing

All structural MRI images underwent standard preprocessing pipeline using the CAT12 toolbox (version 1720, http://www.neuro.uni-jena.de/cat/) based on the SPM12 software (version v7771, https://www.fil.ion.ucl.ac.uk/spm/software/spm12). The CAT12 is an alternative to FreeSurfer, which provides a volume-based approach to reliably estimate multiple surface-based morphological indices without extensive reconstruction of cortical surface. Briefly, data preprocessing included bias-field and noise removal, skull stripping, gray and white matter segmentation (longitudinal segment module), CT estimation and reconstruction of central surface, spherical harmonics-based topology correction, and extraction of FD, GI, and SD. All the steps were conducted in subject native space. After obtaining individual morphological maps (CT, FD, GI, and SD), they were further re-parameterized into a common coordinate system (fsaverage template) and smoothed using a Gaussian kernel.

Morphological index

CT was estimated using a fast and reliable projection-based thickness method (Dahnke et al. 2013). Specifically, following tissue segmentation, a white matter distance map was derived by estimating the distance from the inner gray matter boundary for each GM voxel. Values at the outer GM boundary in the WM distance map (i.e. GM thickness) were then projected back to the inner GM boundary to generate a GM thickness map. FD reflects folding complexity of cortical surface, and was calculated as the slope of a logarithmic plot of surface area versus the maximum *l*-value (Yotter et al. 2011). GI is an indicator of cortical folding, and was calculated as absolute mean curvature (Luders et al. 2006). SD measures the depth of sulci and was calculated as the Euclidean distance between the central surface and its convex hull. Notably, according to the recommendations of the CAT12 manual, individual CT maps were smoothed using a Gaussian kernel with 12-mm full width at half-maximum, whereas the others were smoothed using a Gaussian kernel with 25-mm full width at half-maximum.

Construction of single-subject cerebral morphological networks

We constructed four types of cerebral morphological networks (CT-based networks, CTNs; FD-based networks, FDNs; GI-based networks, GINs; SD-based networks SDNs) for each structural MRI image using the method as described in our previous study (Li et al. 2021). In these networks, nodes represented brain regions and edges represented interregional morphological similarity in the distributions of regional morphology.

Brain parcellation

To define network nodes, we employed three widely used cerebral surface atlases, the DK40 atlas (Desikan et al. 2006), a2009s atlas (Destrieux et al. 2010), and Human Connectome Project Multi-Modal Parcellation (HCP-MMP) atlas (Glasser et al. 2016), which divided the cerebral cortex into 68, 148, and 360 regions of interest (ROIs), respectively. Of note, two ROIs were excluded for the HCP-MMP atlas due to too few vertices in them.

Similarity measure

Similar to our previous study (Li et al. 2021), interregional morphological similarity was estimated with two measures, the KLDs and JSDs. In probability theory and statistics, the KLD is an asymmetric measure of how one probability distribution is different from a second, reference probability distribution. It is a nonnegative value without an upper limit. The JSD is a variant of KLD but with some notable differences, including that it is symmetric and it always has a finite value. First, values of all vertices within each ROI were extracted for each morphological index. Then, a probability density estimate was obtained for each ROI and each morphological index using a MATLAB function (ksdensity). After converting the resultant probability density estimates to probability distribution functions (PDFs), the KLDs between two PDFs *P* and *Q* was computed as follows:

$$KLD(P\|Q) = \sum_{i=1}^{n} P(i)\log\frac{P(i)}{Q(i)}$$

$$KLDs\,(P, Q) = e^{-\left(KLD(P\|Q)+KLD(Q\|P)\right)}$$

where *e* is the natural base, and *n* is the number of sample points (2^8 in this study) (Wang et al. 2016). For the JSDs between *P* and *Q*, the formula is

$$JSD\,(P\|Q) = \frac{1}{2}KLD\left(P\|\frac{1}{2}(P+Q)\right) + \frac{1}{2}KLD\left(Q\|\frac{1}{2}(P+Q)\right)$$

$$JSDs\,(P, Q) = 1 - \sqrt{JSD\,(P\|Q)}$$

The value range for both KLDs and JSDs is [0, 1], with 0 and 1 denoting that two PDFs are completely different and exactly the same, respectively.

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Removal of site effects

For multicentric studies, site effects can be minimized by rigorous study design or alternatively by image post-processing techniques and statistical models. In this study, all structural scans in dataset 1were performed by the same operator according to the fixed operation process in the same type of 3 T MR scanner equipped with the same type of 64-channel head coil. Moreover, the same acquisition parameters were used. Under these strict conditions, no site effects were expected for dataset 1. In contrast, for dataset 2, we utilized a harmonization approach called Combat to moderate site effects on interregional morphological similarities. The Combat harmonization is demonstrated to successfully remove inter-site technical variability while preserving inter-site biological variability in image-based measurements (Fortin et al. 2018). Specifically, the Combat model can be written as follows:

$$y_{ijv} = \alpha_v + X_{ij}\beta_v + \gamma_{iv} + \delta_{iv}\varepsilon_{ijv}$$

where y_{ijv} represents morphological similarity of edge v for subject j in site i, α_v is the average morphological similarity for edge v, X is a design matrix for the covariates of interest (e.g. age, sex or group label), β_v is a vector of regression coefficients corresponding to covariates in X, and ε_{ijv} is the residual term that is assumed to follow a normal distribution with zero mean. The terms γ_{iv} and δ_{iv} represent the additive and multiplicative site effects of site i on edge v, respectively, and are estimated by conditional posterior means as described previously (Johnson et al. 2007). The final ComBat-harmonized morphological similarity for edge v is calculated as follows:

$$y_{ijv}^{ComBat} = \frac{y_{ijv} - \hat{\alpha}_v - X_{ij}\hat{\beta}_v - \gamma_{iv}^*}{\delta_{iv}^*} + \hat{\alpha}_v + X_{ij}\hat{\beta}_v$$

where $\gamma_{iv}^*$and δ_{iv}^* are the empirical Bayes estimates of γ_{iv} and δ_{iv}, respectively.

Network analysis

Thresholding method

Two widely used thresholding methods were adopted to exclude connections with low morphological similarities in each morphological similarity matrix: absolute (similarity-based) and proportional (sparsity-based). For the absolute thresholding method, the same similarity threshold (KLDs or JSDs) was applied to each morphological similarity matrix such that only connections with similarities greater than the threshold were retained. However, this method leads to different numbers of edges across participants and visits in the resulting networks due to differences in the overall similarity strength among the morphological similarity matrices. Thus, the absolute thresholding method allows an examination of the absolute network organization. In contrast, the proportional thresholding method ensured the same number of edges (and thus the same network sparsity, which defined as the ratio of the number of actual edges divided by the maximum possible number of edges in a network) across participants and visits by applying a subject- and visit-specific similarity threshold (KLDs or JSDs) to individual morphological similarity matrices. Therefore, the proportional thresholding method allows an examination of the relative network organization. Network measures derived from these two thresholding methods quantify distinct aspects of brain network organization and together provide a comprehensive way to test brain network alterations in diseases (He et al. 2009). Specifically, given the sparse nature of human brain networks, for the absolute thresholding method the similarity thresholds were determined by generating 20 exponentially spaced points in the range of [0 1], and for the proportional thresholding method the sparsities were determined by generating 20 logarithmically spaced points in the range of [0 1]. These procedures ensured more sampling points for sparser networks, and meanwhile reduced the computational burden.

Network type

For each thresholding method, each morphological similarity matrix $C_{ij} = [c_{ij}]$ was converted into a set of both binarized A_{ij} and weighted W_{ij} cerebral morphological networks:

$$A_{ij} = [a_{ij}] = \begin{cases} 1, \text{if } c_{ij} > \tau \\ 0, \text{otherwise} \end{cases}$$

$$W_{ij} = [w_{ij}] = \begin{cases} c_{ij}, \text{if } c_{ij} > \tau \\ 0, \text{otherwise} \end{cases}$$

where τ is a pre-defined similarity threshold (KLDs or JSDs).

Network parameter calculation

For each binarized and weighted cerebral morphological network, five graph-based network measures were calculated for each node with the GRETNA toolbox (Wang et al. 2015), including clustering coefficient, local efficiency, global efficiency, degree, and betweenness. Details on the calculation and interpretation of the network measures can be found elsewhere (Rubinov and Sporns 2010; Wang et al. 2011). Given that the network measures were calculated as functions of similarity threshold and sparsity, we finally calculated the area under the curve (AUC, i.e. the integral over the entire range of similarity threshold and sparsity, respectively) for each network measure to provide summarized scalars for the TRT reliability analysis.

Inter-site TRT reliability

To provide an overall estimate of inter-site TRT reliabilities of single-subject cerebral morphological networks across all sites, we calculated the ICC for each network measure of each node using a two-way mixed-effects model (ICC (A,1)) (Shrout and Fleiss 1979; McGraw and Wong 1996):

$$ICC(A,1) = \frac{MS_R - MS_E}{MS_R + (k-1)MS_E + \frac{k}{n}(MS_C - MS_E)}$$

where MS_R is the mean square of between-subject variance, MS_E is the mean square of error, MS_C is the mean square of within-subject variance, k denotes the number of repeated observations per participant, and n denotes the number of participants. The ICC approaches 1 for reliable measures and 0 otherwise. For the centers where multiple scans were collected (Center 10 for dataset 1, and Center ATT for dataset 2), only the first one was included. Similar to our previous studies (Wang et al. 2011; Wang et al. 2016; Li et al. 2021), the TRT reliability was categorized as poor (ICC < 0.25), low (0.25 < ICC < 0.4), fair (0.4 < ICC < 0.6), good (0.6 < ICC < 0.75), and excellent (0.75 < ICC < 1).

Effects of different analytical strategies on the inter-site TRT reliabilities of single-subject cerebral morphological networks

In this study, we investigated the inter-site TRT reliabilities of single-subject cerebral morphological networks under different analytical combinations of five factors of morphological index,

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brain parcellation, similarity measure, thresholding method, and network type. To simplify the statistical analysis and provide more realistic recommendations on how to determine the "best" analytical pipeline for single-subject cerebral morphological networks, we examined the effects of these factors one by one instead of putting them into a single complicated statistical model. That is, for a given factor, its effect was examined with the ICC values of all network measures as samples regardless of the choices of the other factors. These statistical analyses were performed for both dataset s 1 and 2.

Effects of morphological index

A one-way repeated ANOVA was performed to test whether different morphological indices (CT, FD, GI, and SD) affected the inter-site TRT reliabilities of single-subject cerebral morphological networks. If a significant main effect was observed, post hoc pairwise comparisons were further performed using paired *t* tests. Multiple comparisons were corrected using the false discovery rate (FDR) procedure at a level of $q < 0.05$.

Effects of brain parcellation

A one-way repeated ANOVA was performed to test whether different parcellation atlases (DK40, a2009s, and HCP-MMP) were associated with different inter-site TRT reliabilities of single-subject cerebral morphological networks. If a significant main effect was observed, post hoc pairwise comparisons were further performed using paired *t* tests. Multiple comparisons were corrected using the FDR procedure at a level of $q < 0.05$. Notably, as the atlases were composed of different numbers of ROIs, the mean ICC values across all ROIs were calculated for each network measure, and used for the statistical analyses.

Effects of similarity measure

A paired-sample *t* test was used to examine the effects of similarity measure (KLDs vs JSDs) on the inter-site TRT reliabilities of single-subject cerebral morphological networks.

Effects of thresholding method

A paired-sample *t* test was used to examine the effects of thresholding method (absolute vs proportional) on the inter-site TRT reliabilities of single-subject cerebral morphological networks.

Effects of network type

A paired-sample *t* test was used to examine the effect of network type (binarized vs weighted) on the inter-site TRT reliabilities of single-subject cerebral morphological networks.

Effects of different scanning protocols on the inter-site TRT reliabilities of single-subject cerebral morphological networks

For dataset 1, additional analyses were performed to examine the effects of different scanning protocols on the inter-site TRT reliabilities of single-subject cerebral morphological networks. Of note, these analyses were restricted to the specific analytical combination that generated the overall highest inter-site TRT reliabilities of single-subject cerebral morphological networks for simplicity.

Effects of data acquisition interval

For dataset 1, the detailed data acquisition intervals were available for the scan-rescans between each pair of sites. This allowed us to test whether longer data acquisition intervals were associated with lower inter-site TRT reliabilities of single-subject cerebral morphological networks. We first screened all pairs of sites with a similar data acquisition interval. These site pairs were further divided into two groups in terms of the data acquisition interval: short-term (mean 11.48 days) and long-term (mean 410 days) (Fig. S1). For a given group of site pairs, the inter-site ICC was calculated between each pair of sites in the group for each network measure of each node using the two-way mixed-effects model, and averaged across all site pairs. For the resulting two sets of ICC values, a paired-sample *t* test was used to examine their differences.

Effects of scanner software version

The MRI scanner software version differed between the Center 9 (Syngo MR VE11C) and the other sites (Syngo MR VD13D) for dataset 1, which enabled us to examine whether different scanner software versions affected inter-site TRT reliabilities of single-subject cerebral morphological networks. First, we chose the Center 8 as a control site since it had minimum time interval from the Center 9 in the data acquisition (1 day). For these two centers, we then separately calculated and averaged their inter-site ICC values with each of the other eight centers for each network measure of each ROI. Finally, a paired-sample *t* test was used to examine the differences in the resulting two sets of ICC values.

Comparisons between inter- and intra-site TRT reliabilities

All participants in dataset 1 were scanned multiple times in the Center 10, allowing us to explore whether inter-site TRT reliabilities were significantly lower than intra-site TRT reliabilities for single-subject cerebral morphological networks. Although the participants in dataset 2 were also scanned multiple times in the center ATT, we found that the released structural MRI images of this center were the same between all visits for each participant. Specifically, to improve comparability, we separately calculated the inter- and intra-site ICC for each network measure of each node as the mean ICC of all pairs of inter- and intra-site scan-rescans, given the different numbers of times between inter-site (10) and intra-site (3) scans. The intra-site ICC was computed using a one-way random-effects model (ICC (1)):

$$ICC(1) = \frac{MS_R - MS_W}{MS_R + (k-1)\,MS_W}$$

where MS_R is the mean square of between-subject variance, MS_W is the mean square of within-subject variance, and k denotes the number of repeated observations per participant. For the resulting two sets of ICC values, a paired-sample *t* tests was used to examine their differences.

However, the above comparison may be biased by differences in the data acquisition interval between inter- and intra-site scan-rescans. To further improve the comparability, we first screened pairs of scan-rescans with a similar data acquisition interval. These site pairs were then divided into two groups in terms of the data acquisition interval: short term (inter-site: mean 4.05 days; intra-site: mean 6.67 days) and long term (inter-site: mean 213.67 days; intra-site: mean 210 days) (Fig. S2). Finally, each group of pairs of scan-rescans were then used to calculate and compare the inter- and intra-site TRT reliabilities of single-subject cerebral morphological networks in the same manner as mentioned above.

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Additional analysis

In addition to the analyses mentioned above, we performed the following analyses to further examine the TRT reliability of single-subject cortical morphological networks.

Effects of harmonization on dataset 1

For dataset 1, no site effects were expected since all structural scans were performed by the same operator according to the fixed operation process in the same type scanner equipped with the same type of head coil, and were collected with the same acquisition parameters. Nevertheless, possible subtle site effects may also exist due to unknown reasons. Thus, we applied the Combat harmonization approach to the cortical morphological networks in dataset 1, and recalculated the network parameters and their inter-site TRT reliabilities under different analytical strategies. The resulting reliabilities were then compared with those derived from the cortical morphological networks in dataset 1 without the harmonization (paired-sample *t* test).

Effects of different analytical strategies on intra-site TRT reliability of single-subject cortical morphological networks

In this study, we examined the effects of five factors (morphological index, brain parcellation, similarity measure, thresholding method, and network type) on the inter-site TRT reliability of single-subject cortical morphological networks. However, such a comprehensive examination is still lacking for intra-site TRT reliability of single-subject cortical morphological networks although several relevant studies exist. Thus, we utilized a new publicly available dataset, the Beijing Normal University test-retest dataset (Lin et al. 2015), to explore the effects of the five factors on the intra-site TRT reliability of single-subject cortical morphological networks. In this dataset, a total of 57 participants (19–30 years) completed two structural image scan sessions within an interval of approximately 6 weeks (40.94 ± 4.51 days). The structural images were processed in the same way as for dataset 1, including preprocessing, morphological index estimation, network construction, network analysis, intra-site ICC calculation, and statistical analysis (i.e. one-way repeated ANOVA for effects of morphological index and brain parcellation, and paired-sample *t* test for effects of the similarity measure, thresholding method, and network type).

Results

Inter-site TRT reliabilities of single-subject cerebral morphological networks.

In this study, we utilized two datasets (datasets 1 and 2) to calculate inter-site TRT reliabilities of five nodal network measures (clustering coefficient, local efficiency, global efficiency, degree, and betweenness) for single-subject cerebral morphological networks derived from four different morphological indices (CT, FD, GI, and FD) under different analytical combinations of brain parcellation (DK40: 68 ROIs; a2009s: 148 ROIs; HCP-MMP: 360 ROIs), similarity measure (KLDs and JSDs), thresholding method (absolute and proportional) and network type (binarized and weighted). These analyses resulted in a total of $5 \times 4 \times (68 + 148 + 358) \times 2 \times 2 \times 2 = 91{,}840$ ICC values for both dataset 1 and dataset 2, which are summarized in Tables 1 and 2, respectively. Out of the values, 77.6% for dataset 1 and 78.5% for dataset 2 fell in the category of fair to excellent TRT reliabilities. Among all network measures under all analytical combinations, Fig. 2 further shows the network measures that exhibited the overall highest and lowest inter-site TRT reliabilities, respectively.

Different analytical strategies affected the inter-site TRT reliabilities of single-subject cerebral morphological networks

Effects of morphological index

A significant main effect of morphological index was observed on the inter-site TRT reliabilities for single-subject cerebral morphological networks (dataset 1: $P < 0.001$; $\eta^2 = 0.272$; dataset 2: $P < 0.001$; $\eta^2 = 0.434$). Post hoc comparisons revealed a common pattern of FD > SD > GI > CT to dataset 1 ($P < 0.05$, FDR-corrected; Cohen's $d_{\text{CT vs FD}} = 1.231$; Cohen's $d_{\text{CT vs GI}} = 0.505$; Cohen's $d_{\text{CT vs SD}} = 1.072$; Cohen's $d_{\text{FD vs GI}} = 0.701$; Cohen's $d_{\text{FD vs SD}} = 0.141$; Cohen's $d_{\text{GI vs SD}} = 0.556$) and dataset 2 ($P < 0.05$, FDR-corrected; Cohen's $d_{\text{CT vs FD}} = 2.064$; Cohen's $d_{\text{CT vs GI}} = 0.795$; Cohen's $d_{\text{CT vs SD}} = 1.784$; Cohen's $d_{\text{FD vs GI}} = 1.250$; Cohen's $d_{\text{FD vs SD}} = 0.212$; Cohen's $d_{\text{GI vs SD}} = 1.001$) (Fig. 3, first column).

Effects of brain parcellation

A significant main effect of brain parcellation was observed on the inter-site TRT reliabilities for single-subject cerebral morphological networks (dataset 1: $P < 0.001$; $\eta^2 = 0.002$; dataset 2: $P < 0.001$; $\eta^2 < 0.001$). Post hoc comparisons revealed a pattern of HCP-MMP > a2009s = DK40 for dataset 1 ($P < 0.05$, FDR-corrected; Cohen's $d_{\text{DK40 vs a2009s}} = 0.051$; Cohen's $d_{\text{DK40 vs HCP-MMP}} = 0.325$; Cohen's $d_{\text{a2009s vs HCP-MMP}} = 0.356$) and HCP-MMP > a2009s > DK40 for dataset 2 ($P < 0.05$, FDR-corrected; Cohen's $d_{\text{DK40 vs a2009s}} = 0.187$; Cohen's $d_{\text{DK40 vs HCP-MMP}} = 0.242$; Cohen's $d_{\text{a2009s vs HCP-MMP}} = 0.202$) (Fig. 3, second column).

Effects of similarity measure

Compared with the JSDs, the KLDs was associated with significantly higher inter-site TRT reliabilities for single-subject cerebral morphological networks (dataset 1: $P < 0.001$; Cohen's $d = 0.217$; dataset 2: $P < 0.001$; Cohen's $d = 0.223$) (Fig. 3, third column). This is in contrast with our expectation, given the mathematical advantages of the JSD relative to KLD and higher intra-site TRT reliabilities of JSDs- than KLDs-based single-subject cortical morphological networks (Li et al. 2021). As thresholding method and network type were two additional factors considered in this study relative to our previous study (Li et al. 2021), we thus separately re-analyzed the effects of similarity measure under each thresholding method (absolute and proportional) and each network type (binarized and weighted) to explore possible reasons for the inconsistency. Significantly higher inter-site TRT reliabilities were observed for the KLDs- than JSDs-based single-subject cortical morphological networks regardless of the choice of network type (dataset 1: $P_{\text{binarized}} < 0.001$, Cohen's $d_{\text{binarized}} = 0.228$; $P_{\text{weighted}} < 0.001$, Cohen's $d_{\text{weighted}} = 0.207$; dataset 2: $P_{\text{binarized}} < 0.001$, Cohen's $d_{\text{binarized}} = 0.256$; $P_{\text{weighted}} < 0.001$, Cohen's $d_{\text{weighted}} = 0.192$) (Fig. 4), whereas opposite patterns were observed when the absolute (KLDs > JSDs; dataset 1: $P < 0.001$; Cohen's $d = 0.321$; dataset 2: $P < 0.001$; Cohen's $d = 0.350$) or proportional (JSDs > KLDs; dataset 1: $P = 0.031$; Cohen's $d = 0.014$; dataset 2: $P < 0.001$; Cohen's $d = 0.082$) thresholding method was used (Fig. 5).

Effects of thresholding method

Compared with the absolute thresholding method, the proportional thresholding method generated significantly higher inter-site TRT reliabilities for single-subject cerebral morphological networks (dataset 1: $P < 0.001$; Cohen's $d = 0.111$; dataset 2: $P < 0.001$; Cohen's $d = 0.115$) (Fig. 3, fourth column).

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Table 1. Summary of TRT reliabilities of single-subject cerebral morphological networks under different analytical strategies for dataset 1.

Morphological index	Parcellation atlas	Similarity measure	Thresholding method	Network type	Clustering coefficient	Local efficiency	Global efficiency	Degree	Betweenness
CT	DK40	KLDs	Absolute	Binarized	0.374 ± 0.278	0.345 ± 0.259	0.591 ± 0.263	0.587 ± 0.264	0.407 ± 0.229
				Weighted	0.366 ± 0.275	0.490 ± 0.248	0.588 ± 0.263	0.586 ± 0.264	0.350 ± 0.226
			Proportional	Binarized	0.292 ± 0.239	0.272 ± 0.252	0.505 ± 0.262	0.518 ± 0.267	0.372 ± 0.217
				Weighted	0.289 ± 0.238	0.298 ± 0.258	0.520 ± 0.256	0.524 ± 0.263	0.312 ± 0.203
		JSDs	Absolute	Binarized	0.321 ± 0.264	0.322 ± 0.256	0.589 ± 0.264	0.588 ± 0.263	0.370 ± 0.226
				Weighted	0.319 ± 0.263	0.475 ± 0.241	0.591 ± 0.263	0.589 ± 0.263	0.338 ± 0.219
			Proportional	Binarized	0.291 ± 0.239	0.273 ± 0.254	0.505 ± 0.262	0.519 ± 0.267	0.373 ± 0.220
				Weighted	0.290 ± 0.238	0.286 ± 0.257	0.514 ± 0.259	0.522 ± 0.267	0.318 ± 0.211
	a2009s	KLDs	Absolute	Binarized	0.277 ± 0.239	0.227 ± 0.212	0.520 ± 0.300	0.525 ± 0.304	0.401 ± 0.256
				Weighted	0.263 ± 0.236	0.282 ± 0.223	0.529 ± 0.306	0.531 ± 0.308	0.323 ± 0.224
			Proportional	Binarized	0.273 ± 0.229	0.217 ± 0.222	0.522 ± 0.304	0.536 ± 0.309	0.391 ± 0.257
				Weighted	0.268 ± 0.227	0.216 ± 0.226	0.522 ± 0.307	0.535 ± 0.310	0.307 ± 0.223
		JSDs	Absolute	Binarized	0.208 ± 0.209	0.239 ± 0.207	0.520 ± 0.300	0.527 ± 0.303	0.372 ± 0.250
				Weighted	**0.205 ± 0.208**	0.300 ± 0.205	0.528 ± 0.304	0.531 ± 0.305	0.316 ± 0.226
			Proportional	Binarized	0.272 ± 0.228	0.216 ± 0.221	0.524 ± 0.304	0.538 ± 0.309	0.391 ± 0.257
				Weighted	0.268 ± 0.227	0.214 ± 0.224	0.524 ± 0.306	0.537 ± 0.310	0.312 ± 0.225
	HCP-MMP	KLDs	Absolute	Binarized	0.426 ± 0.266	0.327 ± 0.242	0.569 ± 0.285	0.575 ± 0.287	0.476 ± 0.265
				Weighted	0.406 ± 0.264	0.421 ± 0.258	0.578 ± 0.289	0.581 ± 0.290	0.383 ± 0.245
			Proportional	Binarized	0.382 ± 0.264	0.317 ± 0.250	0.564 ± 0.296	0.573 ± 0.301	0.468 ± 0.269
				Weighted	0.376 ± 0.261	0.332 ± 0.249	0.566 ± 0.296	0.572 ± 0.301	0.378 ± 0.240
		JSDs	Absolute	Binarized	0.290 ± 0.225	0.258 ± 0.241	0.570 ± 0.285	0.575 ± 0.286	0.439 ± 0.250
				Weighted	0.281 ± 0.222	0.324 ± 0.239	0.577 ± 0.288	0.579 ± 0.288	0.388 ± 0.239
			Proportional	Binarized	0.381 ± 0.263	0.313 ± 0.249	0.565 ± 0.296	0.574 ± 0.301	0.469 ± 0.268
				Weighted	0.376 ± 0.260	0.320 ± 0.246	0.566 ± 0.296	0.574 ± 0.301	0.386 ± 0.243
FD	DK40	KLDs	Absolute	Binarized	0.790 ± 0.269	0.803 ± 0.208	0.942 ± 0.122	0.944 ± 0.145	0.865 ± 0.181
				Weighted	0.787 ± 0.265	0.806 ± 0.214	0.943 ± 0.121	0.942 ± 0.148	0.827 ± 0.200
			Proportional	Binarized	0.796 ± 0.221	0.832 ± 0.183	0.940 ± 0.127	0.938 ± 0.175	0.861 ± 0.166
				Weighted	0.799 ± 0.215	0.832 ± 0.195	0.938 ± 0.131	0.938 ± 0.170	0.811 ± 0.228
		JSDs	Absolute	Binarized	0.743 ± 0.238	0.750 ± 0.253	0.940 ± 0.112	0.946 ± 0.143	0.808 ± 0.210
				Weighted	0.745 ± 0.236	0.756 ± 0.237	0.946 ± 0.117	0.945 ± 0.149	0.786 ± 0.232
			Proportional	Binarized	0.794 ± 0.224	0.830 ± 0.183	0.940 ± 0.132	0.939 ± 0.173	0.857 ± 0.178
				Weighted	0.797 ± 0.219	0.829 ± 0.196	0.937 ± 0.144	0.939 ± 0.170	0.812 ± 0.231
	a2009s	KLDs	Absolute	Binarized	0.809 ± 0.221	0.725 ± 0.266	0.944 ± 0.132	0.950 ± 0.105	0.840 ± 0.189
				Weighted	0.795 ± 0.231	0.763 ± 0.234	0.948 ± 0.117	0.948 ± 0.120	0.774 ± 0.234
			Proportional	Binarized	0.820 ± 0.218	0.778 ± 0.245	0.953 ± 0.129	0.946 ± 0.136	0.857 ± 0.194
				Weighted	0.813 ± 0.225	0.789 ± 0.243	0.950 ± 0.131	0.942 ± 0.152	0.781 ± 0.241
		JSDs	Absolute	Binarized	0.679 ± 0.269	0.659 ± 0.283	0.940 ± 0.132	0.944 ± 0.131	0.794 ± 0.207
				Weighted	0.674 ± 0.273	0.670 ± 0.292	0.946 ± 0.129	0.944 ± 0.142	0.742 ± 0.254
			Proportional	Binarized	0.822 ± 0.216	0.780 ± 0.248	0.954 ± 0.126	0.949 ± 0.130	0.859 ± 0.196
				Weighted	0.818 ± 0.221	0.786 ± 0.249	0.952 ± 0.132	0.948 ± 0.133	0.786 ± 0.245
	HCP-MMP	KLDs	Absolute	Binarized	0.821 ± 0.216	0.739 ± 0.262	0.954 ± 0.112	0.955 ± 0.113	0.830 ± 0.206
				Weighted	0.805 ± 0.227	0.796 ± 0.231	0.957 ± 0.106	**0.958 ± 0.104**	0.776 ± 0.232
			Proportional	Binarized	0.791 ± 0.238	0.745 ± 0.270	0.951 ± 0.128	0.954 ± 0.120	0.840 ± 0.204
				Weighted	0.783 ± 0.246	0.770 ± 0.246	0.955 ± 0.108	0.958 ± 0.104	0.776 ± 0.235
		JSDs	Absolute	Binarized	0.633 ± 0.267	0.616 ± 0.299	0.953 ± 0.115	0.956 ± 0.108	0.765 ± 0.239
				Weighted	0.625 ± 0.269	0.617 ± 0.298	0.956 ± 0.114	0.957 ± 0.109	0.737 ± 0.245
			Proportional	Binarized	0.791 ± 0.241	0.744 ± 0.270	0.952 ± 0.126	0.955 ± 0.113	0.840 ± 0.205
				Weighted	0.784 ± 0.246	0.762 ± 0.253	0.954 ± 0.114	0.957 ± 0.109	0.783 ± 0.235
GI	DK40	KLDs	Absolute	Binarized	0.418 ± 0.272	0.392 ± 0.265	0.723 ± 0.272	0.730 ± 0.274	0.577 ± 0.263
				Weighted	0.405 ± 0.274	0.441 ± 0.273	0.729 ± 0.275	0.733 ± 0.277	0.498 ± 0.250
			Proportional	Binarized	0.373 ± 0.272	0.350 ± 0.279	0.734 ± 0.245	0.749 ± 0.242	0.553 ± 0.281
				Weighted	0.369 ± 0.270	0.374 ± 0.293	0.722 ± 0.261	0.745 ± 0.243	0.486 ± 0.257
		JSDs	Absolute	Binarized	0.344 ± 0.256	0.395 ± 0.252	0.721 ± 0.274	0.729 ± 0.274	0.530 ± 0.263
				Weighted	0.342 ± 0.256	0.455 ± 0.250	0.729 ± 0.275	0.731 ± 0.277	0.483 ± 0.247
			Proportional	Binarized	0.372 ± 0.278	0.353 ± 0.282	0.740 ± 0.245	0.755 ± 0.240	0.558 ± 0.279
				Weighted	0.369 ± 0.277	0.366 ± 0.291	0.733 ± 0.252	0.753 ± 0.239	0.498 ± 0.259
	a2009s	KLDs	Absolute	Binarized	0.465 ± 0.266	0.441 ± 0.280	0.764 ± 0.229	0.772 ± 0.230	0.596 ± 0.279
				Weighted	0.451 ± 0.264	0.452 ± 0.273	0.776 ± 0.228	0.780 ± 0.228	0.489 ± 0.279
			Proportional	Binarized	0.481 ± 0.274	0.464 ± 0.286	0.775 ± 0.230	0.784 ± 0.233	0.590 ± 0.283
				Weighted	0.474 ± 0.273	0.471 ± 0.287	0.776 ± 0.232	0.785 ± 0.233	0.485 ± 0.275
		JSDs	Absolute	Binarized	0.390 ± 0.251	0.439 ± 0.273	0.761 ± 0.235	0.771 ± 0.234	0.568 ± 0.274
				Weighted	0.388 ± 0.252	0.422 ± 0.270	0.772 ± 0.232	0.776 ± 0.232	0.498 ± 0.277
			Proportional	Binarized	0.473 ± 0.273	0.462 ± 0.283	0.774 ± 0.232	0.784 ± 0.235	0.592 ± 0.282
				Weighted	0.469 ± 0.272	0.468 ± 0.285	0.776 ± 0.234	0.785 ± 0.235	0.493 ± 0.278

(continued)

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Table 1. Continued.

Morphological index	Parcellation atlas	Similarity measure	Thresholding method	Network type	Clustering coefficient	Local efficiency	Global efficiency	Degree	Betweenness
	HCP-MMP	KLDs	Absolute	Binarized	0.616 ± 0.285	0.482 ± 0.283	0.781 ± 0.237	0.786 ± 0.233	0.657 ± 0.269
				Weighted	0.592 ± 0.288	0.542 ± 0.289	0.793 ± 0.233	0.794 ± 0.232	0.518 ± 0.278
			Proportional	Binarized	0.557 ± 0.294	0.443 ± 0.295	0.793 ± 0.227	0.803 ± 0.223	0.660 ± 0.272
				Weighted	0.544 ± 0.293	0.466 ± 0.292	0.796 ± 0.226	0.804 ± 0.222	0.520 ± 0.278
		JSDs	Absolute	Binarized	0.368 ± 0.269	0.376 ± 0.275	0.785 ± 0.235	0.789 ± 0.232	0.612 ± 0.266
				Weighted	0.357 ± 0.267	0.394 ± 0.270	0.793 ± 0.231	0.794 ± 0.230	0.520 ± 0.279
			Proportional	Binarized	0.558 ± 0.293	0.443 ± 0.295	0.795 ± 0.224	0.805 ± 0.221	0.660 ± 0.272
				Weighted	0.548 ± 0.293	0.456 ± 0.293	0.798 ± 0.223	0.807 ± 0.219	0.532 ± 0.280
SD	DK40	KLDs	Absolute	Binarized	0.754 ± 0.235	0.693 ± 0.277	0.891 ± 0.191	0.886 ± 0.200	0.772 ± 0.235
				Weighted	0.751 ± 0.236	0.689 ± 0.295	0.890 ± 0.162	0.889 ± 0.174	0.707 ± 0.282
			Proportional	Binarized	0.788 ± 0.194	0.760 ± 0.223	0.866 ± 0.167	0.856 ± 0.205	0.760 ± 0.240
				Weighted	0.786 ± 0.193	0.739 ± 0.250	0.881 ± 0.154	0.857 ± 0.203	0.716 ± 0.264
		JSDs	Absolute	Binarized	0.694 ± 0.244	0.675 ± 0.230	0.887 ± 0.173	0.880 ± 0.199	0.731 ± 0.222
				Weighted	0.692 ± 0.244	0.683 ± 0.237	0.893 ± 0.166	0.890 ± 0.176	0.688 ± 0.277
			Proportional	Binarized	0.784 ± 0.200	0.755 ± 0.223	0.870 ± 0.163	0.862 ± 0.198	0.769 ± 0.224
				Weighted	0.782 ± 0.200	0.742 ± 0.239	0.882 ± 0.143	0.862 ± 0.198	0.714 ± 0.266
	a2009s	KLDs	Absolute	Binarized	0.792 ± 0.227	0.756 ± 0.241	0.910 ± 0.131	0.909 ± 0.140	0.756 ± 0.209
				Weighted	0.788 ± 0.231	0.763 ± 0.234	0.919 ± 0.129	0.919 ± 0.130	0.687 ± 0.261
			Proportional	Binarized	0.795 ± 0.231	0.786 ± 0.228	0.921 ± 0.127	0.925 ± 0.137	0.794 ± 0.189
				Weighted	0.797 ± 0.230	0.782 ± 0.212	0.924 ± 0.138	0.931 ± 0.132	0.698 ± 0.266
		JSDs	Absolute	Binarized	0.691 ± 0.269	0.680 ± 0.274	0.897 ± 0.160	0.902 ± 0.153	0.685 ± 0.239
				Weighted	0.690 ± 0.270	0.695 ± 0.260	0.908 ± 0.150	0.910 ± 0.150	0.627 ± 0.285
			Proportional	Binarized	0.796 ± 0.229	0.784 ± 0.228	0.922 ± 0.128	0.926 ± 0.134	0.794 ± 0.193
				Weighted	0.797 ± 0.228	0.781 ± 0.221	0.925 ± 0.132	0.929 ± 0.138	0.693 ± 0.276
	HCP-MMP	KLDs	Absolute	Binarized	0.815 ± 0.208	0.748 ± 0.248	0.922 ± 0.150	0.923 ± 0.149	0.766 ± 0.236
				Weighted	0.802 ± 0.215	0.806 ± 0.216	0.923 ± 0.157	0.924 ± 0.153	0.692 ± 0.269
			Proportional	Binarized	0.764 ± 0.238	0.700 ± 0.279	0.924 ± 0.145	0.926 ± 0.143	0.780 ± 0.231
				Weighted	0.760 ± 0.238	0.724 ± 0.263	0.926 ± 0.142	0.927 ± 0.144	0.697 ± 0.273
		JSDs	Absolute	Binarized	0.573 ± 0.285	0.517 ± 0.306	0.918 ± 0.151	0.923 ± 0.147	0.682 ± 0.252
				Weighted	0.566 ± 0.288	0.549 ± 0.294	0.924 ± 0.146	0.925 ± 0.147	0.655 ± 0.276
			Proportional	Binarized	0.763 ± 0.240	0.699 ± 0.281	0.925 ± 0.141	0.927 ± 0.140	0.781 ± 0.232
				Weighted	0.759 ± 0.241	0.712 ± 0.271	0.927 ± 0.139	0.928 ± 0.142	0.708 ± 0.270

The network measures showing the overall highest and lowest reliabilities are highlighted in bold font, and presented in Fig. 2. CT, cortical thickness; FD, fractal dimension; GI, gyrification index; SD, sulcal depth; HCP-MMP, Human Connectome Project Multi-Modal Parcellation; KLDs, Kullback–Leibler divergence-based similarity; JSDs, Jensen–Shannon divergence-based similarity.

Effects of network type

Compared with weighted network model, binarized network analysis resulted in significantly higher inter-site TRT reliabilities for single-subject cerebral morphological networks (dataset 1: $P < 0.001$; Cohen's $d = 0.090$; dataset 2: $P < 0.001$; Cohen's $d = 0.234$) (Fig. 3, fifth column).

Different scanning protocols affected the TRT reliabilities of single-subject cerebral morphological networks

Under the specific analytical combination that generated the overall highest inter-site TRT reliabilities (morphological index: FD; brain parcellation: HCP-MMP; similarity measure: JSDs; thresholding method: proportional; network type: binarized), we further examined the effects of different scanning protocols on the TRT reliabilities of single-subject cerebral morphological networks for dataset 1.

Data acquisition interval

The inter-site TRT reliabilities of single-subject cerebral morphological networks were significantly higher for scan-rescans separated by short- than long-term time intervals ($P < 0.001$; Cohen's $d = 0.269$) (Fig. 6).

Scanner software version

Scan-rescans collected with different scanner software versions resulted in significantly lower inter-site TRT reliabilities of single-subject cerebral morphological networks ($P < 0.001$; Cohen's $d = 0.081$) (Fig. 7).

Inter- vs intra-site reliabilities

The inter-site TRT reliabilities were significantly lower than the intra-site TRT reliabilities for the single-subject cerebral morphological networks no matter whether the factor of data acquisition interval was taken into account (all: $P < 0.001$; Cohen's $d = 0.155$; short-term: $P < 0.001$; Cohen's $d = 0.213$; long-term: $P < 0.001$; Cohen's $d = 0.196$) (Fig. 8).

Results of additional analysis

Harmonization significantly improved the inter-site TRT reliabilities of single-subject cerebral morphological networks for dataset 1

After the harmonization of dataset 1, the inter-site TRT reliabilities of cortical morphological networks were significantly improved ($P < 0.001$; Cohen's $d = 0.097$). This finding indicates that site effects cannot be fully excluded via rigorous study design.

Different analytical strategies affected the intra-site TRT reliabilities of single-subject cerebral morphological networks

The intra-site TRT reliabilities of single-subject cortical morphological networks were significantly affected in a similar manner as the inter-site TRT reliabilities by morphological

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Table 2. Summary of TRT reliabilities of single-subject cerebral morphological networks under different analytical strategies for dataset 2.

Morphological index	Parcellation atlas	Similarity measure	Thresholding method	Network type	Clustering coefficient	Local efficiency	Global efficiency	Degree	Betweenness
CT	DK40	KLDs	Absolute	Binarized	0.300 ± 0.153	0.285 ± 0.175	0.469 ± 0.194	0.479 ± 0.192	0.355 ± 0.128
				Weighted	0.294 ± 0.155	0.319 ± 0.151	0.471 ± 0.193	0.477 ± 0.193	0.278 ± 0.122
			Proportional	Binarized	0.250 ± 0.164	0.259 ± 0.188	0.457 ± 0.186	0.469 ± 0.186	0.317 ± 0.130
				Weighted	0.248 ± 0.165	0.266 ± 0.184	0.460 ± 0.187	0.469 ± 0.186	0.250 ± 0.120
		JSDs	Absolute	Binarized	0.265 ± 0.156	0.284 ± 0.179	0.467 ± 0.195	0.480 ± 0.194	0.328 ± 0.132
				Weighted	0.265 ± 0.157	0.315 ± 0.156	0.475 ± 0.194	0.480 ± 0.193	0.276 ± 0.125
			Proportional	Binarized	0.253 ± 0.164	0.263 ± 0.188	0.456 ± 0.186	0.469 ± 0.186	0.317 ± 0.129
				Weighted	0.251 ± 0.164	0.268 ± 0.186	0.459 ± 0.187	0.469 ± 0.186	0.259 ± 0.126
	a2009s	KLDs	Absolute	Binarized	0.311 ± 0.131	0.275 ± 0.138	0.511 ± 0.189	0.518 ± 0.190	0.364 ± 0.145
				Weighted	0.299 ± 0.129	0.348 ± 0.132	0.516 ± 0.185	0.519 ± 0.185	0.266 ± 0.142
			Proportional	Binarized	0.244 ± 0.139	0.231 ± 0.159	0.496 ± 0.184	0.505 ± 0.180	0.357 ± 0.149
				Weighted	0.239 ± 0.139	0.246 ± 0.156	0.501 ± 0.183	0.506 ± 0.179	0.261 ± 0.141
		JSDs	Absolute	Binarized	0.254 ± 0.116	0.281 ± 0.146	0.515 ± 0.186	0.521 ± 0.188	0.354 ± 0.147
				Weighted	0.251 ± 0.116	0.351 ± 0.134	0.521 ± 0.185	0.522 ± 0.186	0.286 ± 0.147
			Proportional	Binarized	0.236 ± 0.139	**0.227 ± 0.158**	0.497 ± 0.184	0.506 ± 0.180	0.365 ± 0.148
				Weighted	0.234 ± 0.139	0.236 ± 0.157	0.500 ± 0.184	0.507 ± 0.179	0.278 ± 0.145
	HCP-MMP	KLDs	Absolute	Binarized	0.367 ± 0.159	0.275 ± 0.156	0.538 ± 0.177	0.547 ± 0.176	0.415 ± 0.148
				Weighted	0.347 ± 0.156	0.322 ± 0.151	0.546 ± 0.175	0.551 ± 0.174	0.280 ± 0.149
			Proportional	Binarized	0.333 ± 0.161	0.267 ± 0.165	0.536 ± 0.177	0.547 ± 0.173	0.418 ± 0.154
				Weighted	0.325 ± 0.159	0.278 ± 0.163	0.541 ± 0.175	0.548 ± 0.171	0.277 ± 0.145
		JSDs	Absolute	Binarized	0.266 ± 0.135	0.282 ± 0.159	0.541 ± 0.176	0.548 ± 0.175	0.416 ± 0.146
				Weighted	0.259 ± 0.134	0.302 ± 0.150	0.548 ± 0.175	0.551 ± 0.174	0.319 ± 0.150
			Proportional	Binarized	0.333 ± 0.161	0.266 ± 0.167	0.537 ± 0.177	0.547 ± 0.172	0.434 ± 0.156
				Weighted	0.327 ± 0.160	0.273 ± 0.166	0.540 ± 0.176	0.548 ± 0.172	0.305 ± 0.151
FD	DK40	KLDs	Absolute	Binarized	0.769 ± 0.125	0.716 ± 0.155	0.930 ± 0.065	0.937 ± 0.060	0.840 ± 0.098
				Weighted	0.758 ± 0.132	0.773 ± 0.123	0.934 ± 0.064	0.938 ± 0.060	0.788 ± 0.146
			Proportional	Binarized	0.740 ± 0.153	0.707 ± 0.186	0.926 ± 0.064	0.933 ± 0.063	0.840 ± 0.115
				Weighted	0.736 ± 0.154	0.722 ± 0.171	0.927 ± 0.064	0.933 ± 0.062	0.787 ± 0.153
		JSDs	Absolute	Binarized	0.682 ± 0.154	0.671 ± 0.159	0.928 ± 0.066	0.936 ± 0.060	0.794 ± 0.117
				Weighted	0.678 ± 0.156	0.728 ± 0.136	0.935 ± 0.062	0.938 ± 0.060	0.769 ± 0.154
			Proportional	Binarized	0.736 ± 0.154	0.704 ± 0.184	0.927 ± 0.064	0.934 ± 0.062	0.840 ± 0.115
				Weighted	0.733 ± 0.155	0.711 ± 0.176	0.928 ± 0.063	0.935 ± 0.061	0.797 ± 0.149
	a2009s	KLDs	Absolute	Binarized	0.763 ± 0.135	0.706 ± 0.152	0.934 ± 0.096	0.938 ± 0.097	0.835 ± 0.110
				Weighted	0.753 ± 0.135	0.727 ± 0.142	0.937 ± 0.098	0.939 ± 0.098	0.769 ± 0.160
			Proportional	Binarized	0.776 ± 0.133	0.742 ± 0.159	0.935 ± 0.095	0.939 ± 0.097	0.836 ± 0.115
				Weighted	0.772 ± 0.134	0.753 ± 0.151	0.935 ± 0.099	0.938 ± 0.099	0.761 ± 0.158
		JSDs	Absolute	Binarized	0.666 ± 0.147	0.661 ± 0.168	0.933 ± 0.096	0.939 ± 0.096	0.782 ± 0.118
				Weighted	0.663 ± 0.147	0.667 ± 0.158	0.938 ± 0.096	0.940 ± 0.096	0.743 ± 0.159
			Proportional	Binarized	0.775 ± 0.132	0.738 ± 0.160	0.936 ± 0.094	0.940 ± 0.095	0.836 ± 0.114
				Weighted	0.773 ± 0.132	0.746 ± 0.155	0.936 ± 0.097	0.940 ± 0.097	0.770 ± 0.160
	HCP-MMP	KLDs	Absolute	Binarized	0.787 ± 0.131	0.685 ± 0.166	0.935 ± 0.066	0.939 ± 0.062	0.824 ± 0.105
				Weighted	0.769 ± 0.137	0.726 ± 0.151	0.939 ± 0.062	0.941 ± 0.060	0.707 ± 0.179
			Proportional	Binarized	0.766 ± 0.150	0.714 ± 0.175	0.939 ± 0.062	0.943 ± 0.058	0.836 ± 0.102
				Weighted	0.759 ± 0.152	0.720 ± 0.169	0.939 ± 0.060	0.943 ± 0.056	0.703 ± 0.174
		JSDs	Absolute	Binarized	0.609 ± 0.172	0.615 ± 0.197	0.935 ± 0.065	0.939 ± 0.062	0.769 ± 0.118
				Weighted	0.602 ± 0.176	0.616 ± 0.186	0.939 ± 0.062	0.940 ± 0.061	0.696 ± 0.173
			Proportional	Binarized	0.767 ± 0.149	0.712 ± 0.175	0.939 ± 0.061	0.943 ± 0.058	0.839 ± 0.101
				Weighted	0.761 ± 0.150	0.716 ± 0.173	0.940 ± 0.060	**0.943 ± 0.056**	0.723 ± 0.167
GI	DK40	KLDs	Absolute	Binarized	0.419 ± 0.144	0.437 ± 0.153	0.722 ± 0.161	0.734 ± 0.162	0.546 ± 0.167
				Weighted	0.412 ± 0.142	0.468 ± 0.147	0.728 ± 0.161	0.735 ± 0.162	0.446 ± 0.195
			Proportional	Binarized	0.374 ± 0.131	0.400 ± 0.175	0.713 ± 0.162	0.727 ± 0.165	0.530 ± 0.169
				Weighted	0.371 ± 0.131	0.414 ± 0.162	0.716 ± 0.160	0.727 ± 0.164	0.435 ± 0.192
		JSDs	Absolute	Binarized	0.376 ± 0.130	0.428 ± 0.151	0.727 ± 0.159	0.739 ± 0.161	0.526 ± 0.166
				Weighted	0.376 ± 0.131	0.469 ± 0.145	0.735 ± 0.160	0.740 ± 0.161	0.449 ± 0.193
			Proportional	Binarized	0.373 ± 0.134	0.402 ± 0.178	0.716 ± 0.161	0.730 ± 0.165	0.529 ± 0.167
				Weighted	0.371 ± 0.134	0.410 ± 0.170	0.718 ± 0.160	0.730 ± 0.164	0.442 ± 0.194
	a2009s	KLDs	Absolute	Binarized	0.443 ± 0.174	0.408 ± 0.164	0.719 ± 0.157	0.731 ± 0.154	0.586 ± 0.162
				Weighted	0.430 ± 0.171	0.435 ± 0.158	0.730 ± 0.148	0.737 ± 0.147	0.469 ± 0.190
			Proportional	Binarized	0.428 ± 0.185	0.397 ± 0.188	0.716 ± 0.157	0.731 ± 0.150	0.577 ± 0.165
				Weighted	0.421 ± 0.185	0.408 ± 0.184	0.722 ± 0.152	0.734 ± 0.145	0.461 ± 0.185
		JSDs	Absolute	Binarized	0.369 ± 0.152	0.417 ± 0.161	0.722 ± 0.156	0.733 ± 0.154	0.566 ± 0.161
				Weighted	0.367 ± 0.151	0.422 ± 0.153	0.732 ± 0.151	0.737 ± 0.151	0.484 ± 0.185
			Proportional	Binarized	0.424 ± 0.184	0.391 ± 0.189	0.717 ± 0.157	0.732 ± 0.150	0.580 ± 0.164
				Weighted	0.419 ± 0.184	0.398 ± 0.187	0.722 ± 0.154	0.734 ± 0.147	0.477 ± 0.186

(continued)

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Table 2. Continued.

Morphological index	Parcellation atlas	Similarity measure	Thresholding method	Network type	Clustering coefficient	Local efficiency	Global efficiency	Degree	Betweenness
	HCP-MMP	KLDs	Absolute	Binarized	0.524 ± 0.181	0.410 ± 0.179	0.725 ± 0.155	0.735 ± 0.151	0.601 ± 0.156
				Weighted	0.497 ± 0.180	0.465 ± 0.182	0.735 ± 0.150	0.741 ± 0.147	0.418 ± 0.184
			Proportional	Binarized	0.473 ± 0.182	0.410 ± 0.192	0.727 ± 0.153	0.740 ± 0.148	0.603 ± 0.156
				Weighted	0.461 ± 0.180	0.422 ± 0.194	0.732 ± 0.150	0.743 ± 0.147	0.414 ± 0.177
		JSDs	Absolute	Binarized	0.380 ± 0.161	0.407 ± 0.185	0.729 ± 0.153	0.738 ± 0.149	0.573 ± 0.154
				Weighted	0.370 ± 0.162	0.435 ± 0.171	0.738 ± 0.149	0.742 ± 0.147	0.453 ± 0.176
			Proportional	Binarized	0.473 ± 0.183	0.407 ± 0.195	0.728 ± 0.153	0.741 ± 0.148	0.618 ± 0.153
				Weighted	0.464 ± 0.182	0.414 ± 0.196	0.732 ± 0.151	0.743 ± 0.147	0.454 ± 0.174
SD	DK40	KLDs	Absolute	Binarized	0.738 ± 0.134	0.692 ± 0.157	0.904 ± 0.104	0.904 ± 0.115	0.797 ± 0.126
				Weighted	0.733 ± 0.139	0.720 ± 0.137	0.905 ± 0.107	0.907 ± 0.110	0.747 ± 0.165
			Proportional	Binarized	0.719 ± 0.151	0.706 ± 0.151	0.893 ± 0.103	0.898 ± 0.103	0.784 ± 0.134
				Weighted	0.718 ± 0.153	0.706 ± 0.160	0.898 ± 0.103	0.901 ± 0.103	0.728 ± 0.176
		JSDs	Absolute	Binarized	0.639 ± 0.160	0.633 ± 0.176	0.902 ± 0.100	0.903 ± 0.112	0.751 ± 0.140
				Weighted	0.639 ± 0.161	0.657 ± 0.162	0.908 ± 0.104	0.907 ± 0.110	0.724 ± 0.181
			Proportional	Binarized	0.717 ± 0.150	0.701 ± 0.153	0.895 ± 0.102	0.899 ± 0.104	0.787 ± 0.136
				Weighted	0.716 ± 0.151	0.700 ± 0.158	0.897 ± 0.106	0.901 ± 0.104	0.744 ± 0.161
	a2009s	KLDs	Absolute	Binarized	0.753 ± 0.150	0.683 ± 0.189	0.895 ± 0.107	0.901 ± 0.102	0.759 ± 0.144
				Weighted	0.743 ± 0.155	0.704 ± 0.174	0.897 ± 0.110	0.901 ± 0.105	0.670 ± 0.187
			Proportional	Binarized	0.751 ± 0.158	0.713 ± 0.196	0.894 ± 0.104	0.897 ± 0.100	0.768 ± 0.145
				Weighted	0.748 ± 0.160	0.720 ± 0.192	0.896 ± 0.107	0.899 ± 0.101	0.673 ± 0.189
		JSDs	Absolute	Binarized	0.632 ± 0.170	0.625 ± 0.186	0.890 ± 0.109	0.900 ± 0.101	0.701 ± 0.149
				Weighted	0.629 ± 0.172	0.631 ± 0.181	0.898 ± 0.106	0.902 ± 0.103	0.653 ± 0.184
			Proportional	Binarized	0.747 ± 0.156	0.705 ± 0.198	0.895 ± 0.102	0.898 ± 0.099	0.767 ± 0.146
				Weighted	0.744 ± 0.158	0.709 ± 0.195	0.897 ± 0.104	0.900 ± 0.099	0.681 ± 0.186
	HCP-MMP	KLDs	Absolute	Binarized	0.764 ± 0.151	0.676 ± 0.179	0.914 ± 0.082	0.917 ± 0.082	0.757 ± 0.151
				Weighted	0.752 ± 0.155	0.716 ± 0.151	0.914 ± 0.081	0.917 ± 0.081	0.635 ± 0.200
			Proportional	Binarized	0.724 ± 0.177	0.696 ± 0.187	0.911 ± 0.083	0.914 ± 0.083	0.767 ± 0.150
				Weighted	0.721 ± 0.178	0.697 ± 0.179	0.912 ± 0.082	0.914 ± 0.083	0.636 ± 0.195
		JSDs	Absolute	Binarized	0.564 ± 0.171	0.558 ± 0.188	0.910 ± 0.084	0.915 ± 0.083	0.680 ± 0.157
				Weighted	0.561 ± 0.172	0.568 ± 0.178	0.913 ± 0.083	0.915 ± 0.082	0.607 ± 0.195
			Proportional	Binarized	0.723 ± 0.176	0.690 ± 0.190	0.912 ± 0.082	0.914 ± 0.083	0.767 ± 0.152
				Weighted	0.720 ± 0.177	0.691 ± 0.185	0.912 ± 0.081	0.914 ± 0.083	0.656 ± 0.190

The network measures showing the overall highest and lowest reliabilities are highlighted in bold font, and presented in Fig. 2. CT, cortical thickness; FD, fractal dimension; GI, gyrification index; SD, sulcal depth; HCP-MMP, Human Connectome Project Multi-Modal Parcellation; KLDs, Kullback–Leibler divergence-based similarity; JSDs, Jensen–Shannon divergence-based similarity.

index (SD > FD > GI > CT; $P < 0.05$, FDR-corrected; Cohen's $d_{CT\ vs\ FD} = 0.723$; Cohen's $d_{CT\ vs\ GI} = 0.149$; Cohen's $d_{CT\ vs\ SD} = 1.078$; Cohen's $d_{FD\ vs\ GI} = 0.587$; Cohen's $d_{FD\ vs\ SD} = 0.408$; Cohen's $d_{GI\ vs\ SD} = 0.905$), brain parcellation (HCP-MMP = a2009s > DK40; $P < 0.05$, FDR-corrected; Cohen's $d_{DK40\ vs\ a2009s} = 0.359$; Cohen's $d_{DK40\ vs\ HCP\text{-}MMP} = 0.259$; Cohen's $d_{a2009s\ vs\ HCP\text{-}MMP} = 0.027$), similarity measure (absolute thresholding method: KLDs > JSDs; $P < 0.001$; Cohen's $d = 0.371$; proportional thresholding method: JSDs > KLDs; $P < 0.001$; Cohen's $d = 0.069$), thresholding method (proportional > absolute; $P < 0.001$; Cohen's $d = 0.073$), and network type (binarized > weighted; $P < 0.001$; Cohen's $d = 0.152$).

Discussion

In this study, we examined the inter-site TRT reliabilities of graph-based network measures derived from single-subject cerebral morphological networks using two multicentric datasets, and systematically evaluated the effects of several key factors on the reliabilities. Generally, we found fair to excellent reliabilities for all network measures regardless of the analytical pipelines. Further, we demonstrated that the reliabilities depended on different analytical choices of morphological index, brain parcellation, similarity measure, thresholding method, and network type, were decreased by longer data acquisition intervals and different scanner software versions, and were significantly lower than those derived from intra-site scan-rescans. These findings suggest single-subject cerebral morphological networks as a reliable method for multicentric studies of human brain networks, and offer recommendations on how to determine analytical pipelines and scanning protocols for improving reliability.

Single-subject cerebral morphological networks are reliable for multicentric studies

In this study, we found generally fair to excellent inter-site TRT reliabilities for the most frequently used graph-based network measures derived from the single-subject cerebral morphological networks regardless of different analytical strategies. These findings are consistent with our previous study on intra-site TRT reliabilities of single-subject cerebral morphological networks (Li et al. 2021). Thus, the single-subject cerebral morphological networks appear to be a reliable approach for both single-site and multicentric studies of human connectome. Nevertheless, we found that the TRT reliabilities of single-subject cortical morphological networks were significantly affected by different analytical strategies and imaging protocols. Particularly, among the factors influencing the TRT reliabilities of single-subject cortical morphological networks, different choices of morphological index play a pivotal role in terms of the effect size.

Folding-based indices generate higher TRT reliabilities of single-subject cerebral morphological networks

We found that different morphological indices were associated with different levels of inter-site TRT reliabilities of the

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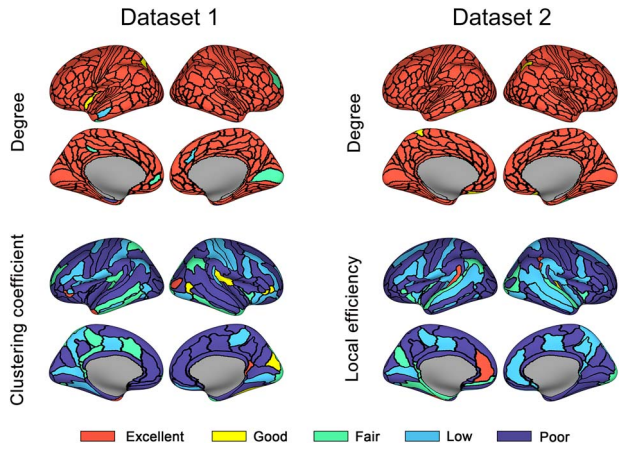


Fig. 2. ICC values of network measures showing the overall highest and lowest inter-site TRT reliabilities among all network measures and analytical combinations. For dataset 1, degree exhibited the overall highest reliabilities (morphological index: FD; parcellation atlas: HCP-MMP; similarity measure: KLDs; thresholding method: absolute; network type: weighted), and clustering coefficient exhibited the overall lowest reliabilities (morphological index: CT; parcellation atlas: a2009s; similarity measure: JSDs; thresholding method: absolute; network type: weighted). For dataset 2, degree exhibited the overall highest reliabilities (morphological index: FD; parcellation atlas: HCP-MMP; similarity measure: JSDs; thresholding method: proportional; network type: weighted), and local efficiency exhibited the overall lowest reliabilities (morphological index: CT; parcellation atlas: a2009s; similarity measure: JSDs; thresholding method: proportional; network type: binarized).

single-subject cerebral morphological networks, with a pattern of FD > SD > GI > CT. This is consistent with our previous finding of intra-site TRT reliabilities of the single-subject cerebral morphological networks (Li et al. 2021). The differences presumably may originate from different computational complexities, sensitivities to noise and/or developmental times and growth rates among the four morphological indexes. For example, CT is a simple reflection of the size and tightness of the arrangement of neurons, glial cells, dendrites, axons and extracellular spaces in the cerebral cortex (Wagstyl and Lerch 2018). In contrast, FD is a highly compact description of the complexity of cortical folding by combining multifactorial information on the folding frequency, sulcus depth, gyral shape, and CT into a single value (Marzi et al. 2020). Such holistic nature of FD may enable it more resistant to noise than those dominated by a single aspect of cortical morphology. This speculation is consistent with a previous study showing higher TRT reliabilities for FD than CT (Madan and Kensinger 2017). With respect to developmental and aging effects, previous studies have shown different trajectories between different morphological indices (Hogstrom et al. 2013; Madan and Kensinger 2016). Theoretically, morphological indices showing faster age-related changes typically have lower TRT reliabilities owing to greater within-subject variance over time. Overall, our current and previous findings together indicate that folding-based morphological indices, in particular FD and SD, provide more TRT reliable representations of single-subject cerebral morphological networks no matter whether the data are collected in a single site or multiple centers. Thus, the FDNs and SDNs are priority recommendations in future studies although different kinds of cerebral morphological networks may provide unique insights into the topological architecture of human brain networks (Li et al. 2021) and uncover distinct alterations caused by brain disorders (Lv et al. 2021).

Higher-resolution brain parcellation is associated with higher TRT reliabilities of single-subject cerebral morphological networks

We found that the HCP-MMP atlas generated the highest TRT reliabilities of single-subject cerebral morphological networks, followed by the a2009s atlas and finally the DK40 atlas. This is in line with our previous study on intra-site TRT reliabilities of the single-subject cerebral morphological networks (Li et al. 2021). These findings jointly indicate that higher-resolution brain parcellation gives rise to more reliable single-subject cerebral morphological networks. Presumably, a major reason for the discrepancy is the more sophisticated methods used to generate the higher-resolution atlases. For example, the HCP-MMP atlas is generated using a multimodal method that incorporates multidimensional information from various imaging modalities (Glasser et al. 2016) wwhereas most structures in the a2005s atlas were defined using a relatively coarse "sulcal" approach via manual tracing (Desikan et al. 2006). Currently, how to generate accurate and reliable parcellation atlases of the human brain is still an open question. In this regard, individualized, multidimensional information guided and phylogeny and ontogeny inspired brain parcellation methods may be promising directions in the future (Eickhoff et al. 2018). According to our findings, the HCP-MMP atlas is the best choice among existing cortical surface atlases for single-subject cerebral morphological network studies from the perspective of TRT reliability.

Different similarity measures affect the TRT reliabilities of single-subject cerebral morphological networks depending on the thresholding methods

We found that the TRT reliabilities of single-subject cerebral morphological networks were influenced by the choices of similarity measure in an interactive manner with the thresholding methods. Specifically, the JSDs yielded significantly higher reliabilities than the KLDs when the proportional thresholding method was employed. This finding is consistent with our previous study on intra-site TRT reliabilities of single-subject cerebral morphological networks (Li et al. 2021). The difference may be due to unique mathematical advantages of the JSD compared with KLD, including that the JSD is symmetric and always has a finite value ([0–1]). In contrast, additional processing steps (e.g. exponential transform) are needed for the KLD to generate symmetric and bounded single-subject cerebral morphological networks. These steps may introduce new noise or lead to unknown disturbance in the estimated interregional morphological similarities, which further reduce the TRT reliabilities. However, the noise or disturbance may more likely play a smoothing role for single-subject cerebral morphological networks with different numbers of edges as a result of the absolute thresholding method, and thus increase the TRT reliabilities. It should be noted that the proportional thresholding method was found to give rise to significantly higher TRT reliabilities of single-subject cerebral morphological networks than the absolute thresholding method. Under this premise, we recommend using the *JSDs* as a priority measure for estimating interregional similarity for single-subject cerebral morphological network studies. In the future, it is warranted to further compare the *JSDs* with other similarity measures, such as earth mover's distance (Sun et al. 2022) and multivariate Euclidean distance (Yu et al. 2018).

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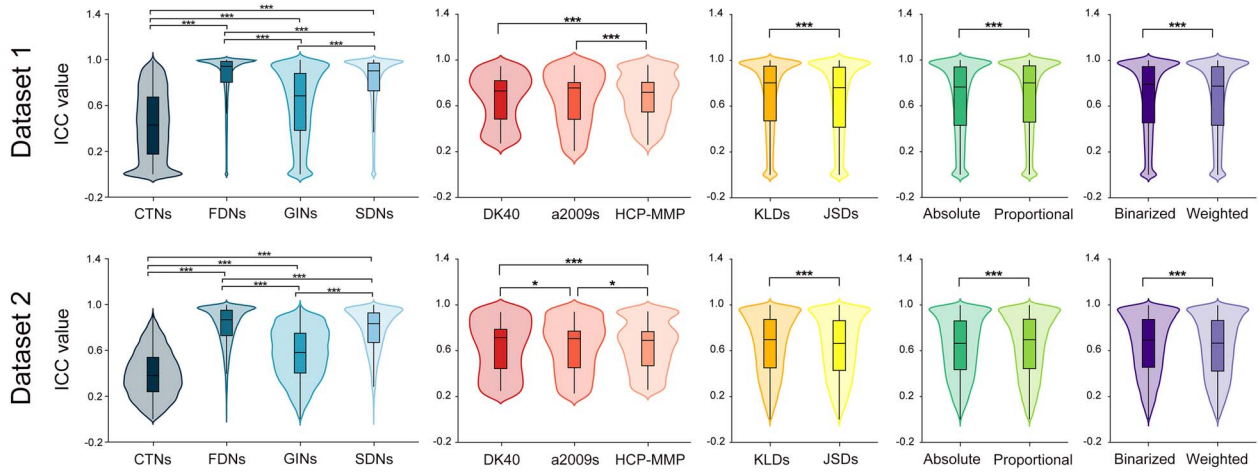


Fig. 3. Effects of different analytical factors on the inter-site TRT reliabilities of single-subject cerebral morphological networks. Different choices of morphological index, brain parcellation, similarity measure, thresholding method, and network type significantly affected the TRT reliabilities of single-subject cerebral morphological networks. ICC, intra-class correlation coefficient; CTNs, cortical thickness networks; FDNs, fractal dimension networks; GINs, gyrification index networks; SDNs, sulcal depth networks; HCP-MMP, Human Connectome Project Multi-Modal Parcellation; KLDs, Kullback–Leibler divergence-based similarity; JSDs, Jensen–Shannon divergence-based similarity.

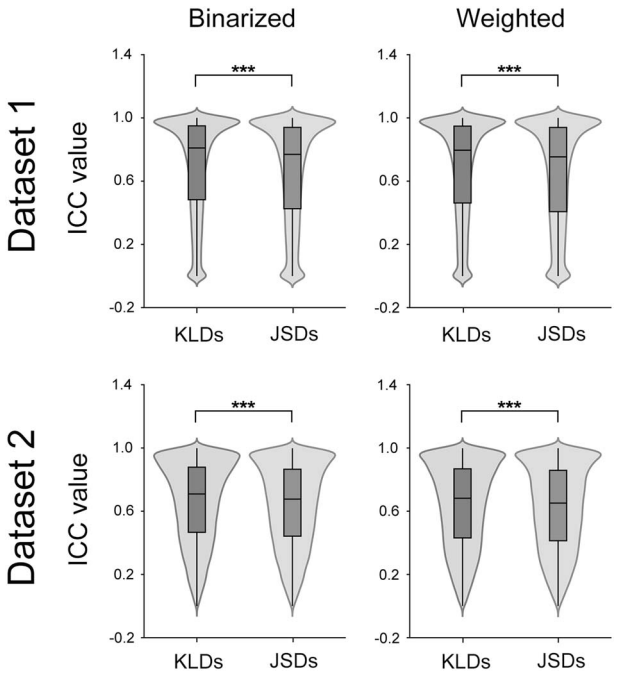


Fig. 4. Effects of similarity measure on the inter-site TRT reliabilities of single-subject cerebral morphological networks under different network types. Significantly higher inter-site TRT reliabilities were observed for the KLDs- than JSDs-based single-subject cortical morphological networks regardless of the choice of network type. ICC, intra-class correlation coefficient; KLDs, Kullback–Leibler divergence-based similarity; JSDs, Jensen–Shannon divergence-based similarity.

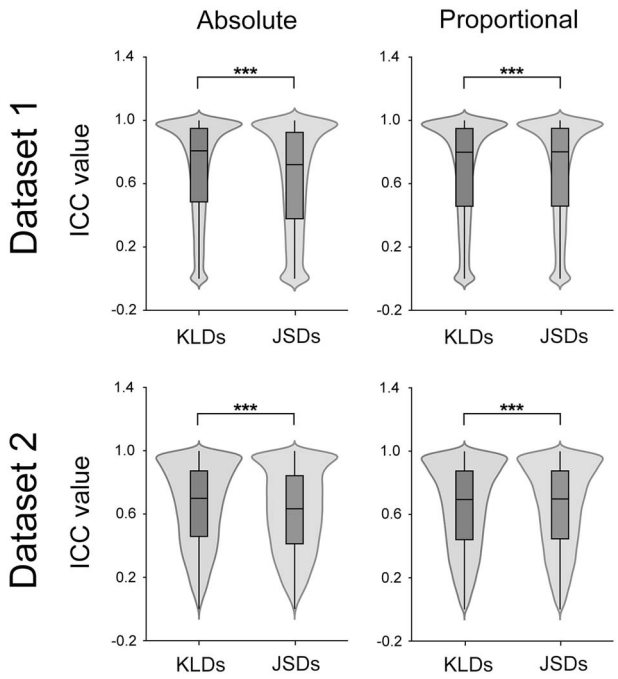


Fig. 5. Effects of similarity measure on the inter-site TRT reliabilities of single-subject cerebral morphological networks under different thresholding methods. Significantly higher inter-site TRT reliabilities were observed for the KLDs-based than JSDs-based single-subject cortical morphological networks when using the absolute thresholding method while opposite patterns were observed when employing the proportional thresholding method. ICC, intra-class correlation coefficient; KLDs, Kullback–Leibler divergence-based similarity; JSDs, Jensen–Shannon divergence-based similarity.

Proportional thresholding method generate higher TRT reliabilities of single-subject cerebral morphological networks

We found that compared with the absolute thresholding method, the proportional thresholding method resulted in significantly higher TRT reliabilities of single-subject cerebral morphological networks. This may be attributable to that the proportional thresholding method corrects for individual differences in the overall connectivity strength, which make the resulting networks more comparable between scans and rescans (i.e. lower within-subject variance). Similar findings were reported in a previous functional network study showing that network measures were more stable across proportional than absolute thresholds (Garrison et al. 2015). Thus, although network measures derived from both the thresholding methods quantify distinct aspects of brain network organization and together provide a comprehensive way to test brain network alterations in diseases (He et al. 2009), we recommend using the proportional thresholding method for future studies of single-subject cerebral morphological

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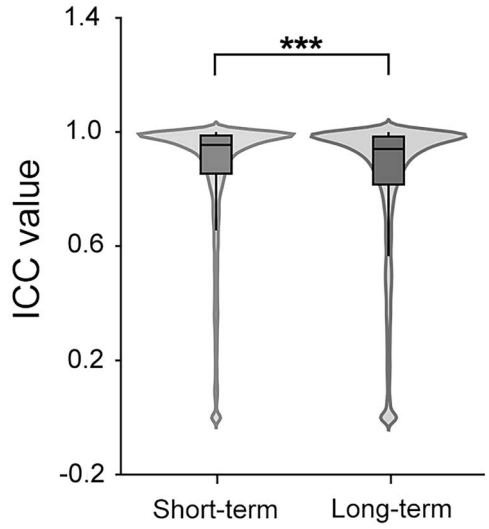


Fig. 6. Effects of data acquisition interval on the inter-site TRT reliabilities of single-subject cerebral morphological networks. Significantly higher inter-site TRT reliabilities were observed for scan-rescans separated by short- than long-term intervals. ICC, intra-class correlation coefficient.

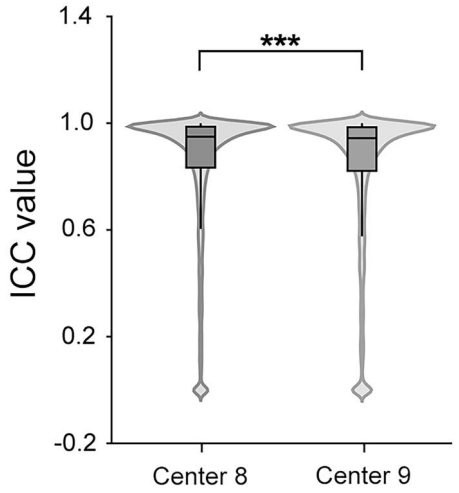


Fig. 7. Effects of scanner software version on the inter-site TRT reliabilities of single-subject cerebral morphological networks. Different scanner software versions significantly reduced the inter-site TRT reliabilities of single-subject cerebral morphological networks. ICC, intra-class correlation coefficient.

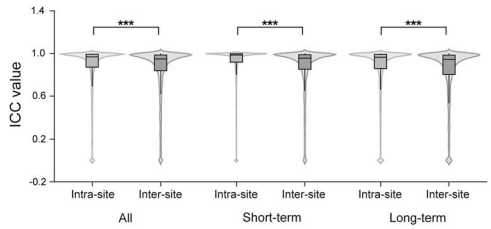


Fig. 8. Differences between inter-site and intra-site TRT reliabilities of single-subject cerebral morphological networks. Significantly higher inter-site than intra-site TRT reliabilities were observed for the single-subject cerebral morphological networks no matter whether the factor of data acquisition interval was taken into account. ICC, intra-class correlation coefficient.

networks under circumstances where TRT reliability is the leading consideration (e.g. studies aiming to establish diagnostic and prognostic biomarkers for diseases). It should be noted that although we demonstrated the higher priority of the proportional thresholding method than the absolute one in terms of the TRT reliability, it is still an open question regarding what specific sparsity should be used for single-subject cerebral morphological networks. Insights into this issue can be gained by examining the TRT reliabilities of single-subject cerebral morphological networks under other thresholding methods, such as mixture modeling (Bielczyk et al. 2018), efficiency cost optimization (De Vico et al. 2017), and minimum spanning three (Tewarie et al. 2015).

Binarized networks yield higher TRT reliabilities of single-subject cerebral morphological networks

We found that binarized network model outperformed weighted network model in reliably charactering single-subject cerebral morphological networks derived from multicentric datasets. This is consistent with our previous studies on intra-site TRT reliabilities of both functional and single-subject cerebral morphological networks (Wang et al. 2011; Wang et al. 2016). Compared with weighted networks, binarized networks are simple for subsequent calculation and interpretation of graph-based network measures, and thus are widely used in previous brain network studies. However, binarized networks may lead to an insufficiently accurate characterization of the topological organization of human brain networks due to the neglect of interregional connectivity strengths. In contrast, weighted networks are thought to topologically characterize human brain networks more accurately by modeling interregional connectivity strengths (Cole et al. 2010), and thus may be more sensitive to capture subtle network alterations under conditions where interregional connectivity strengths alter while the wiring backbone is intact. Nevertheless, it should be emphasized that high accuracy does not mean good reliability. Indeed, accumulating evidence from functional and structural brain networks studies have indicated that neuroimaging derived interregional connectivity strengths fluctuate with different levels of artifact noise, such as in-scanner head motion (Van Dijk et al. 2012; Baum et al. 2018). Analogously, in-scanner head motion was recently found to increase interregional morphological covariance and significantly alter the topological organization of morphological covariance networks (Pardoe and Martin 2022). Based on these previous findings, we speculate that the lower TRT reliabilities of weighted single-subject cerebral morphological networks may be due to disturbances caused by artifact noise in interregional morphological similarities, which lead to more within-subject variance and thus lower TRT reliabilities. According to our discussion, the choice of network type is a trade-off between multiple factors including accuracy, reliability, interpretability and computation cost. At this stage, binarized networks are recommended for studying single-subject cerebral morphological networks in future research.

Longer data acquisition intervals are associated with lower TRT reliabilities of single-subject cerebral morphological networks

We found that long-term data acquisition intervals significantly reduced the reliabilities of nodal centrality measures of single-subject cerebral morphological networks. This finding is consistent with previous functional brain network studies showing lower long- than short-term intra-site reliabilities (Zuo et al. 2012; Jin et al. 2018). Lower reliabilities for longer data acquisition intervals are intuitive to understand in the context of normal aging changes. For example, CT is found to continually decrease across the entire age range of 4.1–88.5 years old (Fjell et al. 2015). Thus,

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longer data acquisition intervals are associated with greater aging changes, which in turn result in more within-subject variance and finally lower reliabilities. In line with this interpretation, a recent study reported lower reliabilities for morphological estimates from scans with longer data acquisition intervals (Hedges et al. 2022). The relatively unreliable morphological estimates due to long-term data acquisition intervals may further lead to the lower reliabilities of single-subject cerebral morphological networks as observed in this study. In this regard, researchers should carefully determine the time intervals for longitudinal follow-up studies to obtain reliable results for single-subject cerebral morphological networks.

Different scanner software versions reduce TRT reliabilities of single-subject cerebral morphological networks

We found that different scanner software versions caused significant decreases in the reliabilities of single-subject cerebral morphological networks. A recent study found that an upgrade from a Siemens MAGNETOM 3 T Tim Trio system to a Siemens MAGNETOM 3 T Prisma Fit system led to significant alterations in morphological estimates (Plitman et al. 2021). Presumably, the alterations may be due to differences in image quality (Kruggel et al. 2010; Vollmar et al. 2010). These findings jointly indicate that differences in scanner system and software are an important confounding factor for studies on structural MRI-derived morphological features and networks. It is interesting to study whether this factor affects other modalities of brain networks in the future.

Intra-site data collection is associated with higher TRT reliabilities of single-subject cerebral morphological networks

For multicentric MRI studies, a variety of factors may affect the reliabilities, including scanner hardware (e.g. scanner manufacturer, field strength and transmitter/receiver coils), software, imaging parameters (e.g. voxel size, echo time and repetition time), and signal-to-noise ratio (Sasaki et al. 2008; Teipel et al. 2011; Fox et al. 2012; Seo et al. 2012; Shinohara et al. 2017; Tong et al. 2019). Thus, larger variabilities and lower reliabilities were observed for inter- than intra-site scan-rescans (Vollmar et al. 2010; Teipel et al. 2011; Kurokawa et al. 2021). Here we further showed that even the data were collected with the same type of hardware, the same version of software (excluding Center 9) and the same acquisition process (dataset 1), inter-site reliabilities were still lower than intra-site reliabilities for single-subject cerebral morphological networks. These findings suggest the existence of additional unknown factors that have negative effects on the reliabilities of multicentric MRI studies. Previous studies have shown that different environments affect brain activity (Norwood et al. 2019). A recent study further showed that weather and season had significant effects on brain volume (Book et al. 2021). Accordingly, we speculate that environmental differences due to changes in geography between sites may be a major source of the observed lower inter- than intra-site reliabilities of single-subject cerebral morphological networks. More insights into our findings may benefit from future studies that systematically explore environmental effects on brain imaging phenotypes (Liu et al. 2023). It should be noted that the harmonization seems to be a necessary step for multicentric studies since it further improved the inter-site TRT reliabilities of single-subject cortical morphological networks even in the context of rigorous study design.

Limitations and future directions

This study has several limitations. First, although the two multicentric datasets used in this study gave rise to largely comparable findings, the samples sizes were small. Moreover, all structural images were collected from 3 T MRI scanners. From a practical point of view, MRI data included in clinical multicentric studies (particularly retrospective) are frequently from scanners with large variability in various aspects, such as field strengths. Thus, more comprehensive evaluation is needed in the future to evaluate the multicentric TRT reliabilities of single-subject cerebral morphological networks. Second, accumulating evidence indicates that in-scanner head motion causes systemic bias in morphological estimates from structural MRI images, such as gray matter volume and CT (Reuter et al. 2015; Alexander-Bloch et al. 2016; Pardoe et al. 2016; Savalia et al. 2017). The systemic bias is further demonstrated in a recent study to subsequently increase morphological covariance between regions and finally lead to significant alterations in topological organization of morphological covariance networks (Pardoe and Martin 2022). Following these findings, we speculate that single-subject cerebral morphological networks and their reliabilities will also be affected by in-scanner head motion, which warrant future research. Third, only four morphological indexes, for which our single-subject cerebral morphological network approaches were established previously, were used to study multicentric TRT reliabilities. Future studies are required to extend our approaches to other morphological indexes (e.g. surface area and surface normal) and further examine their both intra- and inter-site reliabilities. Finally, it should be noted that there are numerous approaches currently available to construct single-subject morphological brain networks. Since these approaches differ largely in the rationale and hypothesis, it is not clear whether our findings hold true for other approaches. Thus, future studies are needed to test the generalization of our findings.

Conclusion

In conclusion, this study demonstrated high multicentric TRT reliabilities of graph-based network measures derived from single-subject cerebral morphological networks, which can be further improved by optimizing analytical strategies and scanning protocols. These findings indicate that single-subject cerebral morphological networks can serve as a reliable approach for characterizing large-scale brain networks in multicentric studies.

Author Contributions

J.W. designed the study; G.Y., T.L., S.J., N.W., J.L., and C.W. analyzed data; G.Y. wrote the manuscript; J.W. and H.H. revised the manuscript.

Supplementary material

Supplementary material is available at *Cerebral Cortex* online.

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Data Availability

All data that support the findings of this study are from publicly available datasets.

CRediT author statement

Guole Yin (Data curation, Methodology, Software, Validation, Visualization, Writing—original draft), Li Ting (Data curation, Methodology, Software), Suhui Jin (Data curation, Formal analysis, Validation), Ningkai Wang (Data curation, Formal analysis), Junle Li (Data curation, Formal analysis), Changwen Wu (Formal analysis), Hongjian He (Writing—review and editing), Jinhui Wang (Conceptualization, Writing—review and editing).

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